

Coverage-Aware Conservative Q-Learning for Offline Reinforcement Learning

Adapting Pessimism to Dataset Density
IE540 Term Project

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June 2026

Main Project Takeaway:

Offline RL should be highly conservative where the dataset is silent, but structurally less pessimistic where empirical data is abundant.

2. The High-Stakes Nature of Offline RL

The Setting: The agent must learn an improved policy *exclusively* from a fixed, historically logged dataset $\mathcal{D} = \{(s, a, r, s')\}$. No new environment interactions are permitted.

Online RL (Standard)

- ▶ Interacts dynamically with environment.
- ▶ Uses trial-and-error to observe rewards.
- ▶ Agent easily discovers and corrects initialization mistakes.

Offline RL (Constrained)

- ▶ Crucial when trial-and-error is risky or unethical (Robotics, Healthcare).
- ▶ **Core Challenge:** Distribution shift between the behavior policy π_β and the learned policy π_θ .

3. Tabular Gridworld MDP Formulation

To isolate and study the offline distribution-shift mechanism away from deep network noise, we formalize a controlled MDP:

- ▶ **State Space ($\mathcal{S}$):** 7×7 discrete grid (49 states) with 6 internal obstacles.
- ▶ **Action Space ($\mathcal{A}$):** 4 deterministic actions (Up, Down, Left, Right).
- ▶ **Reward Function (R):**
 - ▶ -1 per normal step (encourages shortest path).
 - ▶ -2 for collisions with walls or obstacles.
 - ▶ 0 upon reaching the terminal Goal state.
- ▶ **Dataset Constraint:** The algorithm can only sample from a pre-collected batch. Online rollouts are strictly forbidden during value iteration.

4. Core Failure Mode: Out-of-Distribution (OOD) Extrapolation

Standard Q-learning fails catastrophically in the offline setting due to the $\max_{a'} Q(s', a')$ operator in the Bellman backup.

Dataset-Supported Actions

- ▶ Actions seen frequently in $\mathcal{D}$.
- ▶ Targets are grounded in true environmental rewards.
- ▶ Q-values are statistically reliable.

Unsupported Actions (OOD)

- ▶ Unseen actions may have falsely optimistic Q-values due to initialization errors.
- ▶ The greedy policy **exploits** these errors, trapping the agent in loops or obstacles.

5. Base Paper: Conservative Q-Learning (CQL)

Reference: *Conservative Q-Learning for Offline Reinforcement Learning* (Kumar et al., 2020)

Theoretical Foundation:

- ▶ Instead of constraining the policy directly (like BCQ or BRAC), CQL alters the optimization landscape of the critic.
- ▶ **Pessimism as Protection:** It learns a conservative, lower-bound estimate of the true Q-value.
- ▶ It explicitly suppresses the values of actions not present in the dataset, ensuring the agent "does not trust" optimistic unknowns.
- ▶ **Result:** CQL is theoretically motivated to learn conservative/lower-bound value estimates under its assumptions.

6. The Standard CQL Objective

The base algorithm appends a regularization penalty to the temporal difference error:

CQL Objective Function

$$L(Q) = L_{\text{Bellman}} + \alpha \left[\log \sum_a \exp Q(s, a) - Q(s, a_{\text{data}}) \right]$$

Deconstructing the Equation:

- ▶ L_{Bellman} : The standard DP update ensuring long-term reward propagation.
- ▶ α : The fixed hyperparameter weight dictating the severity of pessimism.
- ▶ $\log \sum \exp Q$: Pushes down values for *all* possible actions.
- ▶ $-Q(s, a_{\text{data}})$: Counteracts the push-down for actions actively observed in $\mathcal{D}$.

7. Open Question: The Limitation of Fixed Alpha

The Structural Flaw: Standard CQL applies the exact same penalty scale (α) everywhere, acting as a global, uniform blanket.

Rarely-Covered States

High uncertainty.

Requires stronger pessimism to prevent the policy from chasing OOD hallucinations into obstacles.

Well-Covered States

Low uncertainty.

Risk: High pessimism here actively distorts valid gradients, blocking the agent from taking optimal safe paths.

Research Question: Can we mitigate over-pessimism by dynamically mapping the penalty coefficient to local dataset coverage?

8. Own Extension: Coverage-Aware CQL

My Proposition: Replace the static hyperparameter α with a state-dependent function $\alpha(s)$ that scales inversely with dataset visitation counts $N(s)$.

Square-Root Inverse Frequency Penalty

$$\alpha(s) = \text{clip} \left(\alpha_0 \cdot \sqrt{\frac{\bar{N}}{N(s) + 1}}, 0.05, 1.20 \right)$$

- ▶ $\bar{N}$: The mean visitation count across the entire state space.
- ▶ **Square-Root Scaling:** Inspired by UCB confidence bounds. It prevents the penalty from decaying too rapidly after just a few observations.
- ▶ **Clipping:** Ensures extreme outliers do not cause numerical instability.

9. Algorithmic Workflow

The experiment strictly adheres to the offline RL evaluation protocol:

1. **Dataset Collection:** Log historical trajectories using a behavior policy. Halt all environment interaction.
2. **Density Estimation:** Pass over the dataset to compute state visitation frequencies $N(s)$ and the global mean $\bar{N}$.
3. **Offline Value Iteration:** Train the tabular Q-function / Q-table using the adaptive $\alpha(s)$ augmented Bellman objective.
4. **Zero-Shot Evaluation:** Extract the final greedy policy $\pi(s) = \arg \max_a Q(s, a)$ and deploy it in the environment to measure true task success.

10. Compared Methods & Experimental Rigor

To ensure a fair test of the distribution shift mechanism, all methods evaluate against the **exact same 8 random dataset seeds**.

Method	Pessimism Rule
Offline Q-learning	No penalty ($\alpha = 0$)
Fixed-alpha CQL	Uniform, static penalty ($\alpha = 0.50$)
Coverage-aware CQL	Adaptive penalty ($\alpha_0 = 0.50$)

Global Hyperparameters: Discount factor $\gamma = 0.97$, Learning rate = 0.05, Training epochs = 80.

11. Dataset Coverage & The "Narrow Corridor" Trap

Behavior Policy Profile: A mostly-good heuristic policy injected with mild exploration noise. We generate 50 offline episodes per seed.

100%

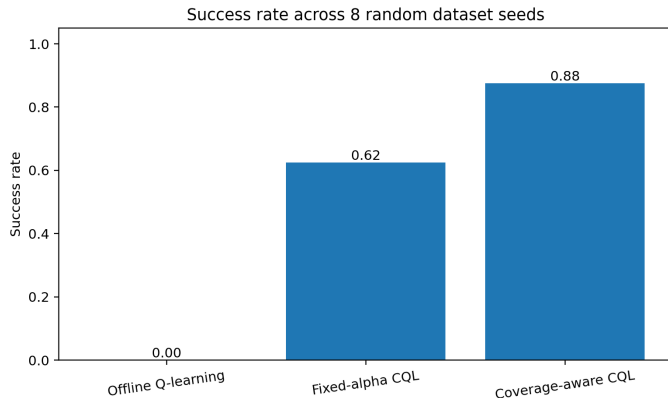
Behavior Policy Success

16.6% ($\pm 1.6\%$)

State-Action Coverage

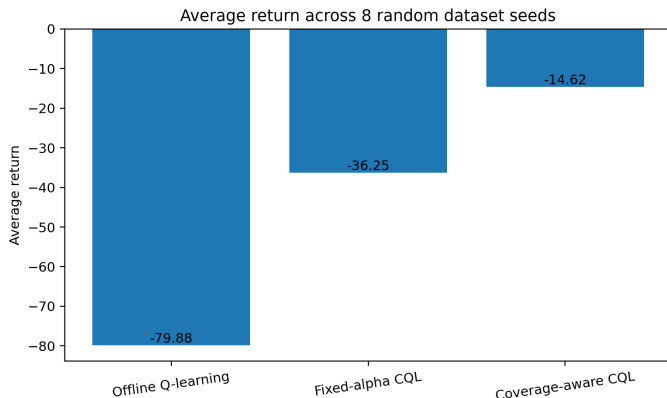
The Trap: Even though the dataset contains perfect paths to the goal, over 80% of the MDP is completely unsupported. Standard algorithms will fail here because the uninitialized voids look mathematically better than the true negative step costs.

12. Empirical Result 1: Task Success Rate



Coverage-aware CQL achieves the highest reliability across 8 variance seeds.

13. Empirical Result 2: Average Trajectory Return



Higher is better (fewer step costs). CA-CQL minimizes meandering path costs.

14. Analytical Results Breakdown

Method	Success Rate	Average Return
Offline Q-learning	0.0%	-79.88 ± 0.35
Fixed-alpha CQL	62.5%	-36.25 ± 34.85
Coverage-aware CQL	87.5%	-14.62 ± 10.25

Why does Coverage-Aware CQL perform best in this toy setting?

- ▶ **Offline Q** completely succumbs to OOD traps (0% success).
- ▶ **Fixed CQL** escapes the traps, but uniform pessimism suppresses the true value gradients inside the safe corridor. This leads to erratic, meandering paths (high variance: ± 34.85).
- ▶ **CA-CQL** maintains the defensive wall around the corridor, but lowers the penalty inside it, resulting in higher success rate, shorter episodes, and better average return. (± 10.25).

15. Discussion & Methodological Limitations

What this controlled experiment shows:

Dataset coverage $N(s)$ is a useful low-cost signal for tuning conservatism in this tabular offline setting.

Limitations & Future Work:

- ▶ **Scale:** This is a controlled tabular isolation study, not a large-scale deep continuous benchmark (like D4RL).
- ▶ **Continuous Adaptation:** Exact state counts $N(s)$ are impossible in continuous pixel or joint-torque spaces. Scaling requires density estimation models (e.g., Variational Autoencoders or Normalizing Flows).
- ▶ **Granularity:** Evaluating state-action density $N(s, a)$ is theoretically superior to pure state density $N(s)$.

16. Conclusion & References

Final Conclusion: CQL effectively addresses offline RL overestimation using pessimistic value learning. Modulating this pessimism via coverage-aware $\alpha(s)$ reduces OOD errors without punishing well-supported data regions.

References:

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- ▶ Fujimoto, S., & Gu, S. (2021). A Minimalist Approach to Offline RL (TD3+BC).

Thank You – Questions?